Shivashant Manohar

AI/ML Engineer | retrieval, fine-tuning and evaluation, measured against real baselines

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Education

VIT-AP University Amaravati, Andhra Pradesh — 2023 - 2027
Bachelor of Technology (B.Tech) in Computer Science and Engineering (AI and ML) | CGPA 8.05/10
Class XII, NBIPU Bengaluru: 94% | Class X, Rashtrotthana Vidya Kendra Dharwad: CGPA 9.2

Experience

Open Source Contributor Agent Orchestrator by Untrivial-ai (12.6k GitHub stars), Remote — Sep 2026 - Present

- Opened PR #5638 (+577 code and +694 test lines, under review) that shows each coding agent's context-window usage in the session model, CLI and doctor check, so exhausted agents can be respawned.
- Opened PR #5956 (under review) that treats an update-check timeout as a network failure instead of a crash.

Projects

NODUS Company memory that flags plans shaped like past outages, Graph Hacks 2026

18/18 reworded risky plans caught, 0/18 false alarms; TF-IDF baseline: 18/18 false alarms

- Turned 5 public post-mortems (Cloudflare, CrowdStrike, Knight Capital, GitLab, AWS S3) into role-and-relationship patterns in a FalkorDB graph, each condition citing a source sentence verified word for word.
- Wrote a deterministic compiler from lessons to parameterised Cypher, so no LLM-written query runs and plans match by structure, not wording; it names the one missing step that would complete a repeat (20/20).
- Declared the evaluation before any test data existed; shipped a live Next.js + FastAPI demo with a private graph per visitor and cut the slowest page from about 26 s to 4 s.

Python, FalkorDB, Cypher, Knowledge Graphs, FastAPI, Next.js | github.com/hey-shiv/nodus

Cover Song Retrieval Two-stage music search on the Da-TACOS benchmark

MAP 0.018 to 0.122 (6.8x) on 13,000 queries; 38 ms/query, within 1.11x of exhaustive search

- Built a PyTorch TCN encoder (supervised contrastive loss) that shortlists 30 of 15,000 recordings, then re-ranks them with key-rotated subsequence dynamic time warping written from the textbook recurrence.
- Reported the first benchmark run as a negative result (4.5x behind), traced it to 2.1% shortlist recall and "hub" tracks, and fixed each cause with settings chosen only on held-out data.
- Made training bit-for-bit reproducible from one seed, with 142 tests and a draft paper generated from the result files.

Python, PyTorch, NumPy, SciPy, Metric Learning, Dynamic Time Warping, pytest | github.com/hey-shiv/cover-version-retrieval

AgentAlign-Lab Verifier-guided fine-tuning of a terminal coding agent, no human raters

Pass rate 7.1% to 11.3% (+4.2 pts, 95% CI +1.2 to +7.7); unsafe runs 41% to 27%

- Built 140 sandboxed terminal tasks with deterministic verifiers (pytest, exact JSON, safety rules) that score every step Qwen2.5-Coder-1.5B takes, then tested on 42 unseen tasks.
- Diagnosed a null first run: training prompts did not match what the agent sees at inference, and only 11 of 80 "chosen" runs had succeeded.
- Rebuilt it from 1,296 rollouts as SFT on clean successful steps plus step-level DPO on pairs whose rejected action a check proves bad; QLoRA on one free Kaggle T4.

Python, PyTorch, Hugging Face Transformers, TRL, PEFT, QLoRA, DPO | github.com/hey-shiv/AgentAlign-Lab

Hulu Support Agent Local-LLM support agent on real Twitter customer-support data

Macro-F1 0.606 (95% CI 0.513 to 0.673) on 200 hand labels; keyword 0.369, TF-IDF 0.355

- Classifies messages into 10 intents, drafts replies grounded in the 3 most similar past resolutions, and escalates to a human with a reason code, all on local models (Qwen2.5 7B, Llama 3.1 8B judge).
- Found and fixed a retrieval leak where test messages retrieved themselves, and a labelling audit that corrected 101 data-entry errors; both are covered by regression tests.

Python, Ollama, RAG, Embeddings, scikit-learn, pandas | github.com/hey-shiv/hulu-support-agent

Skills

Languages: Python, TypeScript, JavaScript, Go, SQL
Machine Learning: PyTorch, TensorFlow, Keras, scikit-learn, NumPy, Deep Learning, NLP, Metric Learning
LLMs: Hugging Face Transformers, PEFT, TRL, QLoRA, DPO, RAG, Embeddings, Ollama
Tools: Git, GitHub Actions, pytest, Linux, Docker, Google Cloud